VYOMA — The Offline AI Receptionist

She sees you, listens to you, answers from our school's own knowledge, and informs the staff — with **no internet at all**.

💯 100% OFFLINE

🔒 DATA STAYS HERE

💸 ₹0 RUNNING COST

1. SEES
Camera detects a face — Vyoma greets the visitor

OpenCV

→

2. LISTENS
Records speech until you pause, converts it to text

Whisper (STT)

→

3. THINKS
A small AI brain reads your question **+ FACTS** retrieved from our knowledge base

Qwen 2.5 LLM + RAG

→

4. SPEAKS
The reply is turned into a natural voice

Piper (TTS)

→

5. ACTS
Logs the visit & notifies the staff member on their phone

SQLite + tools

📚 Open-Book Exam (RAG)

- Our website & documents become its "textbook"
- Each question fetches the 3 most relevant facts
- Vyoma answers **only** from those facts — no making things up!

🛠️ Agentic = Takes Action

- Registers the visitor's name & company
- Logs every visit with purpose & time
- Sends "your guest is waiting" alerts to staff

🌍 Why Offline Matters

- **Privacy:** voices & faces never leave this laptop
- **Cost:** no subscriptions, no cloud API fees
- **Reliability:** works even where internet doesn't

⚙️ Built With

Ears	faster-whisper "small" (CPU)
Brain	Qwen 2.5 · 3B · 4-bit · Ollama (GPU)
Memory	nomic-embed-text + cosine similarity
Voice	Piper (English + Hindi voices)
Eyes	OpenCV Haar cascade
Diary	SQLite database
Code	Python · runs on a 4 GB GPU laptop

🏫 Where It Can Work

- School & college receptions
- Hospital enquiry desks
- Government help counters
- Libraries & museums
- Small offices — after hours too!

🤝 Helper, not a replacement

- Handles repeated questions, like an ATM handles routine cash — humans handle the rest

🚀 What's Next

- Animated face with lip-sync on screen
- Full Malayalam & Hindi conversation
- Faster replies, directional microphone

🧪 Try It!

- Ask: "Tell me about our school's achievements"
- Ask: "I'm here to meet the Principal"
- Watch us switch the Wi-Fi OFF — it still works!

VYOMA · named after the Sanskrit word for "sky" · inspired by ISRO's Vyommitra · built with free & open-source software · Class 8 Science Fest Project